1. catkin_make

To open a catkin code as a project, use “Open File or Project” and select the top-level CMakeLists.txt of the catkin workspace (e.g. “catkin_ws/src/CMakeLists.txt“). Select the catkin build folder (e.g. “catkin_ws/build“) as the build directory and ‘Run CMake‘ (in order to enable debugging add following line into arguments edit box:

```
-DCMAKE_BUILD_TYPE=Debug).
```

Recently this has started to fail with errors like “CMake Error: The source directory “/opt/ ros/lunar/share/catkin/cmake” does not appear to contain CMakeLists.txt.”, because the main CMakeLists is a symlink to a non-writable location. The workaround is to make a copy of toplevel.cmake instead of using a symlink:

```
mv CMakeLists.txt CMakeLists.txt.old

cp /opt/ros/lunar/share/catkin/cmake/toplevel.cmake CMakeLists.txt
```

To be able to modify all the files in the workspace add those lines in “src/CMakeLists.txt” :

```
#Add custom (non compiling) targets so launch scripts and python files show up in QT Creator’s project view.

file(GLOB_RECURSE EXTRA_FILES */*)

add_custom_target(${PROJECT_NAME}_OTHER_FILES ALL WORKING_DIRECTORY $ {PROJECT_SOURCE_DIR} SOURCES ${EXTRA_FILES})
```

If you want the project to be named something else than “Project” then add a line at the top with project(!MyProjectName).

You may specify the correct catkin devel and install spaces at Projects->Build Settings by providing the following CMake arguments:

```
-DCATKIN_DEVEL_PREFIX=../devel - DCMAKE_INSTALL_PREFIX=../install
```

2. catkin tools

With the new catkin_tools, there is no longer a top-level make file for the whole workspace. Instead, open each package as an individual project in QtCreator. Make sure, that the build folder is set to ws/build/your_package instead of ws/build.